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Proceedings, Part IV

Part 4

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Breast Cancer Classification Based on Fuzzy Rules and Deep Learning Techniques

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Abstract. Cancer is a common disease and affects most patients, early diagnosis and treatment help increase the chance of recovery. Mammography is one of the popular screening methods today, in the paper we propose a method to classify cancer on mammograms using fuzzy rules combined with deep learning models. Features from the machine learning model are considered and adjusted through the fuzzy inference system to achieve better classification efficiency. The results of the study achieved over 94.83% proving the effectiveness of the proposed method. These results provide a crucial foundation for our ongoing development of fuzzy learning techniques to classify and identify tumors in mammograms.

Keywords: Breast cancer · Fuzzy rules · Deep learning

1 Overview

In both emerging and developing countries, breast cancer ranks as the most lethal disease affecting women. According to the World Health Organization (WHO), it is projected that there will be 19.3 million new diagnoses of breast cancer by the year 2025 [2]. Early detection and accurate classification of breast cancer can enable patients to access suitable treatment options. This form of cancer is the most commonly diagnosed among women, impacting 2.1 million individuals annually and accounting for a significant proportion of cancer-related deaths in the female population. Early detection of breast cancer before surgery and accurate identification of the type of breast cancer can reduce the rate of surgical treatment and minimize mortality. Mammograms are used to diagnose abnormalities such as masses, calcifications, architecture distortions, or area asymmetries. Due to the superposition nature of 2D mammograms, different views of a

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breast could be needed for richer information. Standard mammography views are the craniocaudal (CC) view and the mediolateral oblique (MLO) view for both breasts, which are taken directly from the above and from an angled side of the breast, respectively. Extra views might also be generated depending on practical needs. Digital breast tomosynthesis (DBT), also regarded as 3D mammography, has been introduced to provide more spatial context for detailed examination of the breasts and is emerging as the standard of breast imaging care [1].

Cheng et al. [3] introduced a novel method that uses fuzzy logic to enhance the contrast of mammograms. The goal is to improve the image quality, helping diagnosticians detect abnormalities in breast tissue more accurately. The proposed method has proven effective in enhancing the contrast of mammograms, helping to highlight important details and improve the detection of tumors or lesions. Experimental results show that this method outperforms traditional enhancement techniques, especially in handling low-contrast images. Jiang et al. [4] proposed a new method combining fuzzy logic and structure tensor to improve the contrast of mammograms. The goal is to enhance the image quality to support more effective breast cancer diagnosis. Baharuddin et al. [5] developed a BIRADS (Breast Imaging Reporting and Data System) classification model based on Type-2 fuzzy logic, using data from mammography reports. BIRADS is a standardized system for preparing mammography reports, aiming to reduce confusion in the image evaluation process. However, BIRADS classification can be challenging due to inter-individual and intra-individual variability. Therefore, this study aims to use Type-2 fuzzy logic to handle these uncertainties.

2 Background

2.1 VGG19

VGG-19 [6] is a deep convolutional neural network consisting of 19 weight layers—16 convolutional layers and 3 fully connected layers. Its straightforward, repetitive architecture makes it easy to understand and implement (Fig. 1):

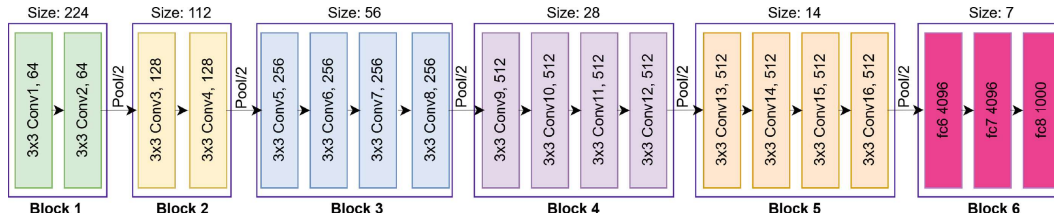


Fig. 1. An illustration of VGG19 network architecture [6].

2.2 Resnet50

ResNet [7] is the most popular architecture used at the moment. ResNet is also the earliest architecture to apply batch normalization. Although it is a very deep network with the number of layers up to 152, thanks to the application of special techniques that we will learn below, the size of ResNet50 is only about 26 million parameters. The architecture with few parameters but the effectiveness of ResNet brought victory in the ImageNet competition in 2015 (Fig. 2).

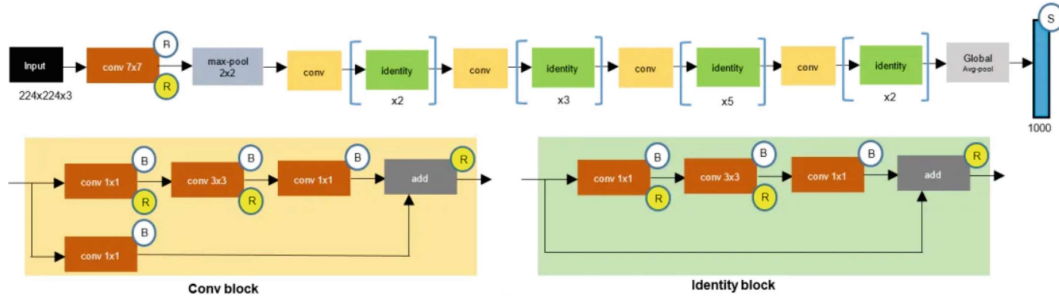


Fig. 2. Resnet network architecture [7].

ResNet-50 [7] is a deep convolutional neural network architecture developed by Kaiming He and colleagues in 2015. It belongs to the ResNet (Residual Network) family, a family of models designed to address the problem of performance degradation as the depth of the network increases. ResNet-50 specifically has 50 convolutional layers and uses residual connections to help information flow through the network efficiently.

2.3 Rule-Based Algorithm

- **JRIP**: is an implementation of the RIPPER (Repeated Incremental Pruning to Produce Error Reduction) algorithm [10], a popular rule-based classification algorithm. It is designed to generate efficient and understandable rule sets for classification problems. The rules are generated in an iterative process and refined through pruning to reduce the error on the test set. Once completed, JRIP prioritizes the rules, with the first matching rule being applied first.
- **PART**: is a hybrid algorithm between decision trees [11] and rule-based methods. The main goal is to build rules from parts of a decision tree, instead of building the entire tree as usual. PART generates rules by building a partial decision tree on the data set. Each rule is extracted from a branch of the tree and then the cases covered by that rule are removed from the data set. Continue the process on the remaining data until there are no cases left.

2.4 Fuzzy Logic

Fuzzy logic [12] is a technique for processing information and making decisions in the presence of ambiguous or uncertain values. In contrast to classical logic,

Algorithm 1. RIPPER Algorithm**Require:** Training dataset M with classes $\{T_1, T_2, \dots, T_k\}$ **Ensure:** Rule set R

- 1: Sort the classes in ascending order of size: $T_1, T_2, \dots, T_k$
- 2: Initialize the rule set $R \leftarrow \emptyset$
- 3: **for** each class T_i in $\{T_1, T_2, \dots, T_k\}$ **do**
- 4: **repeat**
- 5: Find the best rule R_i that covers instances of class T_i
- 6: Optimize the rule R_i using IREP
- 7: Add R_i to the rule set R
- 8: Remove instances covered by R_i from M
- 9: **until** no instances of class T_i remain uncovered
- 10: **end for**
- 11: Create a default rule for instances not covered by any rule in R
- 12: **return** R

Algorithm 2. PART Algorithm**Require:** Training dataset M **Ensure:** Rule set R

- 1: Initialize the rule set $R \leftarrow \emptyset$
- 2: Initialize the dataset $M_{\text{remaining}} \leftarrow M$
- 3: **while** $M_{\text{remaining}}$ is not empty **do**
- 4: Build a partial decision tree using $M_{\text{remaining}}$
- 5: Extract the best leaf (rule) from the partial decision tree
- 6: Add the extracted rule to R
- 7: Remove instances covered by the rule from $M_{\text{remaining}}$
- 8: Discard the rest of the partial decision tree
- 9: **end while**
- 10: **return** R

which operates solely with binary values (0 or 1), fuzzy logic accommodates a range of values between 0 and 1, thus enabling the representation of states that are neither entirely true nor entirely false. The integration of fuzzy learning methodologies into machine learning is referred to as fuzzy machine learning. These models prove particularly advantageous in managing uncertain information commonly encountered in medical imaging, enhancing the inference and analysis of abnormal regions within images, and consequently improving the efficiency of diagnosis and classification. A fuzzy inference system facilitates decision-making through the application of fuzzy rules and a comprehensive knowledge base (Fig. 3).

The fuzzy logic process handles input and output variables through three phases: fuzzification, fuzzy inference, and defuzzification. During fuzzification, crisp values are converted into fuzzy values by mapping actual inputs to membership functions and determining each input's membership grade. Fuzzy membership functions help to convert the crisp values of the input into fuzzy values, which represent the degree of belonging of an element to a fuzzy set. The mem-

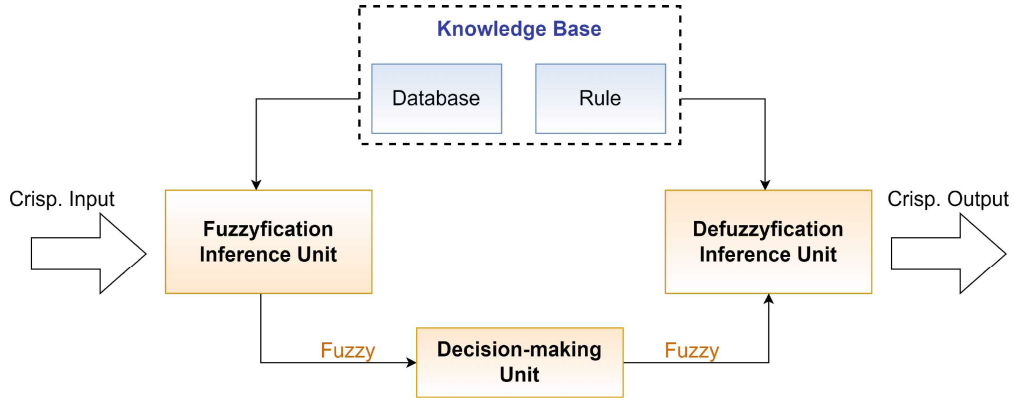


Fig. 3. Fuzzy inference system (FIS) [8].

bership function of a fuzzy set M is defined by $\delta_M(x)$, where $\delta_M(x) \in [0, 1]$ represents the degree of membership of x to the fuzzy set M . Each membership function is shown graphically to illustrate how much each input participates in a set. The triangular membership function, used in this paper, offers mathematical simplicity. It is defined by three parameters $\{a, b, c\}$, where the membership function $\delta_M(x)$ for each value x is described in Eq. 1. These parameter values are derived from expert knowledge.

$$\delta_M(x) = \begin{cases} 0 & \text{if } x \leq a \\ \frac{x-a}{b-a} & \text{if } a < x \leq b \\ \frac{c-x}{c-b} & \text{if } b < x \leq c \\ 0 & \text{if } x > c \end{cases} \quad (1)$$

The second phase involves fuzzy inference, where the system processes input values. Using IF-THEN linguistic rules from a knowledge base, the fuzzy inference system draws conclusions. These rules apply input membership values as weights to influence the fuzzy output sets and final conclusions. In the final phase, defuzzification converts the fuzzy results back into precise, crisp values. This defuzzification process takes a weighted average of the x values, where the weights are the membership function values $\delta(x)$.

$$Centroid = \frac{\int x \cdot \delta(x) dx}{\int \delta(x) dx} \quad (2)$$

where $\delta(x)$ is the membership function (fuzzy attribute level) of the value x ; x is a continuous input variable.

2.5 Evaluation Metrics

To evaluate the object detection model, AP and mAP metrics are often used. These metrics are determined through the following values: Precision is the correct prediction which is the ratio of correct Positive classification - TP (True

positive) on the total number of predicted cases with FP (False positive) being the wrong prediction calculated by Eq. 3; Recall is the sensitivity calculated by the ratio of correct Positive classification - TP (True positive) on the total number of Positive cases with FN (False negative) being the missed prediction calculated by Eq. 4. Since Recall and Precision are two different concepts, we have the F1_score index which allows us to evaluate the average between Precision and Recall.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

$$\text{F1_score} = 2 \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

To evaluate the average precision across all classes, we use the mAP measure [9]. The mAP measure is calculated using Eq. 6.

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (6)$$

The loss metric is used to measure the degree of deviation between the actual value and the predicted value of the model. In multiclass classification problems, the goal of the training process is to minimize this loss value. The loss function used in the paper is given by the Eq. 7:

$$\text{Loss} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(p_{i,c}) \quad (7)$$

where N : number of samples, C : number of classes (number of labels), $y_{i,c}$: the actual label of the i sample, which is a one-hot value (1 if the sample belongs to class c , otherwise it is 0) and $p_{i,c}$: the predicted probability of class c for sample i (output of softmax).

3 Proposed Method

In this paper, we propose a three-stage general model. The stages include: (1) preprocessing to create input data for a deep learning model, (2) feature extraction and training, and (3) classification using fuzzy rule sets. The proposed model is presented in Fig. 4.

- **Pre-processing data:** The input image will be normalized, stored in .jpg format and sized at 224×224 . Image pre-processing is performed on the input data by normalizing the image size and enhancing the image using histogram algorithms.

- **Feature extraction and convert to fuzzy rules:** The processed images are passed through network models to extract features. The features after the

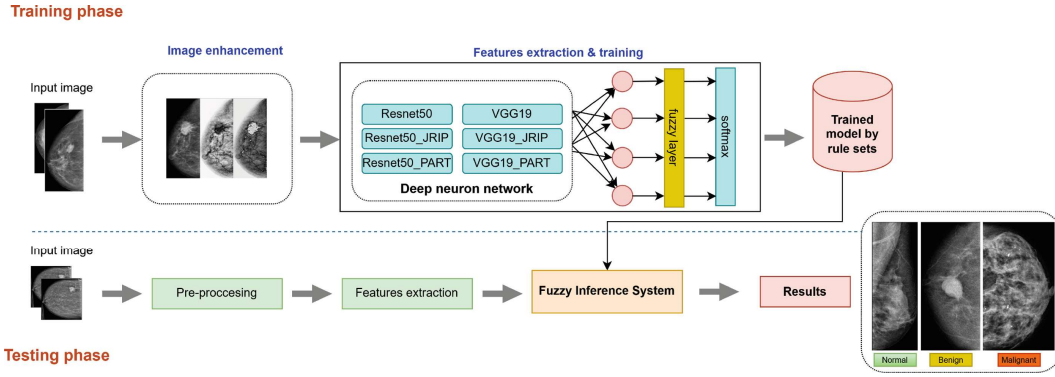


Fig. 4. The proposed general model for breast cancer classification based on fuzzy rule sets and deep learning techniques.

process are extracted into rule sets using the JRIP and PART algorithms presented in Sect. 2.3. Convert the threshold values in the rule set into fuzzy membership functions. The fuzzy dependency function in the paper presented in Sect. 2.4.

- **Classification using FIS:** The Fuzzy Inference System (FIS) model classification algorithm begins with training the system based on Resnet and VGG19 models, where the features extracted from these network models are used to construct fuzzy rules. These fuzzy rules will combine the fuzzy input values with the corresponding classification results, and determine the fuzzy membership

Algorithm 3. Classification using fuzzy rules.

Input: Trained Model T , Test Features X_{test} , Fuzzy Inference System FIS

Output: Predicted Class C

- 1: Normalize X_{test}
 - 2: Initialize Fuzzy values: Fuzzy_values $\leftarrow []$
 - 3: **for** each feature x_i in X_{test} **do**
 - 4: **for** each fuzzy set A_k in FIS **do**
 - 5: Compute membership degree $\delta_{M_k}(x_i)$ using membership function (using Eq.1)
 - 6: Add $\delta_{M_k}(x_i)$ to Fuzzy_Values
 - 7: **end for**
 - 8: **end for**
 - 9: Initialize rule activations: Rule_activations $\leftarrow []$
 - 10: **for** each rule R in FIS **do**
 - 11: Compute activation strength of R using Fuzzy_values and fuzzy operators
 - 12: Add activation strength to Rule_activations
 - 13: **end for**
 - 14: Aggregate Rule_activations to compute crisp scores S_c for each class
 - 15: Defuzzify scores using Centroid method (using Eq. 2)
 - 16: Predict class $C \leftarrow \arg \max S_c$
 - 17: **return** C
-

Table 1. Rule sets were created from the models.

Algorithm	Rule sets
Resnet50_JRIP	Rule 1: If feature_13 ≥ -2.769054 AND feature_189 ≤ 0.388679 AND feature_254 ≥ 0.370445 AND feature_180 ≤ -0.085201 AND feature_31 ≥ -0.353784 AND feature_374 ≥ -0.074123 AND feature_170 ≤ 2.332747 AND feature_307 ≥ -1.639286, then class = 0 Rule 2: If feature_14 ≤ -1.571163 AND feature_272 ≤ -0.234635 AND feature_103 ≤ -1.555267 AND feature_155 ≥ 0.600437, then class = 0 ... Rule 43: If feature_10 ≥ 1.710023 AND feature_234 ≥ 0.125292 AND feature_317 ≤ -0.299676 AND feature_23 ≥ 0.149888 AND feature_9 ≤ 5.152686, then class = 1 Rule 44: If feature_325 ≤ -0.427231 AND feature_30 ≤ -9.245857, then class = 1
Resnet50_PART	Rule 1: If feature_49 > 19.049036 AND feature_249 > -0.780902, then class = 1 Rule 2: If feature_181 ≤ -8.234502 AND feature_401 > -0.94244, then class = 1 ... Rule 107: If feature_380 ≤ 0.323817 AND feature_136 ≤ 0.807774, then class = 0 Rule 108: If feature_39 ≤ -0.172926, then class = 2
VGG19_JRIP	Rule 1: If feature_227 ≤ 0.347213 AND feature_320 ≤ -1.739756 AND feature_813 ≤ -0.425521 AND feature_39 ≤ 2.182981 AND feature_159 ≥ -0.871579, then class = 0 Rule 2: If feature_85 ≤ 0.700167 AND feature_120 ≥ 1.482226 AND feature_266 ≤ -0.639873 AND feature_56 ≤ -4.336328 AND feature_687 ≥ -1.341465, then class = 0 ... Rule 53: If feature_889 ≤ -0.98143 AND feature_367 ≤ -3.344875 AND feature_144 ≤ -1.367553, then class = 1 Rule 54: If feature_724 ≥ 3.234887 AND feature_44 ≥ 8.010759, then class = 1
VGG19_PART	Rule 1: If feature_24 > 29.375553, then class = 2 Rule 2: If feature_41 > 24.75745 AND feature_280 ≤ 3.498626, then class = 1 ... Rule 106: If feature_7 > -9.389853 AND feature_182 > 0.532455, then class = 2 Rule 107: If feature_6 ≤ -6.632608, then class = 1

Table 2. The fuzzy rules of VGG19_PART is generated from the rule sets of table 1.

Algorithm	Fuzzy rules
VGG19_PART	Rule 1: If feature_24 is High, then class = 2 Rule 2: If feature_41 is High AND feature_280 is Low or Medium, then class = 1 Rule 3: If feature_4 is Medium or High AND feature_22 is High AND feature_626 is Medium or High AND feature_205 is Medium or High AND feature_236 is Low or Medium, then class = 1 Rule 4: If feature_4 is Medium or High AND feature_60 is High AND feature_178 is Low or Medium, then class = 1 ... Rule 106: If feature_7 is Medium or High AND feature_182 is Medium or High, then class = 2 Rule 107: If feature_6 is Low or Medium, then class = 1

function for each feature. After training, the system will be used to classify the test images. For the test images, the features will be extracted and converted into fuzzy values. The FIS system will apply the learned fuzzy rules to calculate the fuzzy outputs based on the features of the test images. Then, the defuzzification process will be performed, converting the fuzzy values into deterministic

values through the weighted average method. Finally, the classification results of the test image will be compared with the classes in the trained model, thereby determining the class of the test image based on the obtained deterministic values. This process allows the system to accurately classify test images based on previously trained fuzzy features (Tables 1 and 2).

4 Experimental Results

4.1 Applicable Scenarios

For the proposed model, we conduct experiments under six scenarios with the following training parameters (Table 3):

Table 3. Recommended scenarios and training parameters.

Scenario	Name	Learning rate	Epoch	Batch size	Num classes	Num rules
1	Resnet50	0.0001	100	32	3	-
2	Resnet50_JRIP	0.0001	100	32	3	44
3	Resnet50_PART	0.0001	100	32	3	108
4	VGG19	0.0001	100	32	3	-
5	VGG19_JRIP	0.0001	100	32	3	54
6	VGG19_PART	0.0001	100	32	3	107

In scenarios 2,3,5,6 we use algorithms to generate rule sets from the data extracted from the network models. The goal of this process is to improve the accuracy of the model by better handling ambiguous or contradictory cases in the data set. After conducting experiments with various sets of parameters, we selected identical parameters for all scenarios. The learning rate is set at 0.0001, with the training being conducted 100 times. The number of classes (Num classes) being 3, corresponding to benign, malignant and normal.

4.2 Installation Environment and Dataset

Installation Environment: The system runs in a Python environment on Kaggle, using an Intel Xeon 2.00 GHz processor with 30GB RAM. It also utilizes a graphics processor with a 16 GB RAM configuration and the Nvidia Tesla P100 GPU. Tensorflow v1.5 is the library used for network model training.

-Experimental Dataset: The dataset is compiled from the VinDr dataset, supplemented with data from several hospitals. The number of images we use in the paper includes 242 benign images, 310 malignant images and 493 normal images. The dataset is divided in the ratio of 70:30.

4.3 Training Results

In this section, we present the results from our experiments. Including the measurements of Loss, Accuracy, Confusion Matrix and Precision, Recall, F1-score based on the equation presented in Sect. 2.5 for the proposed scenarios, training and testing times, as well as the detection and classification results for the proposed scenarios (Fig. 5).

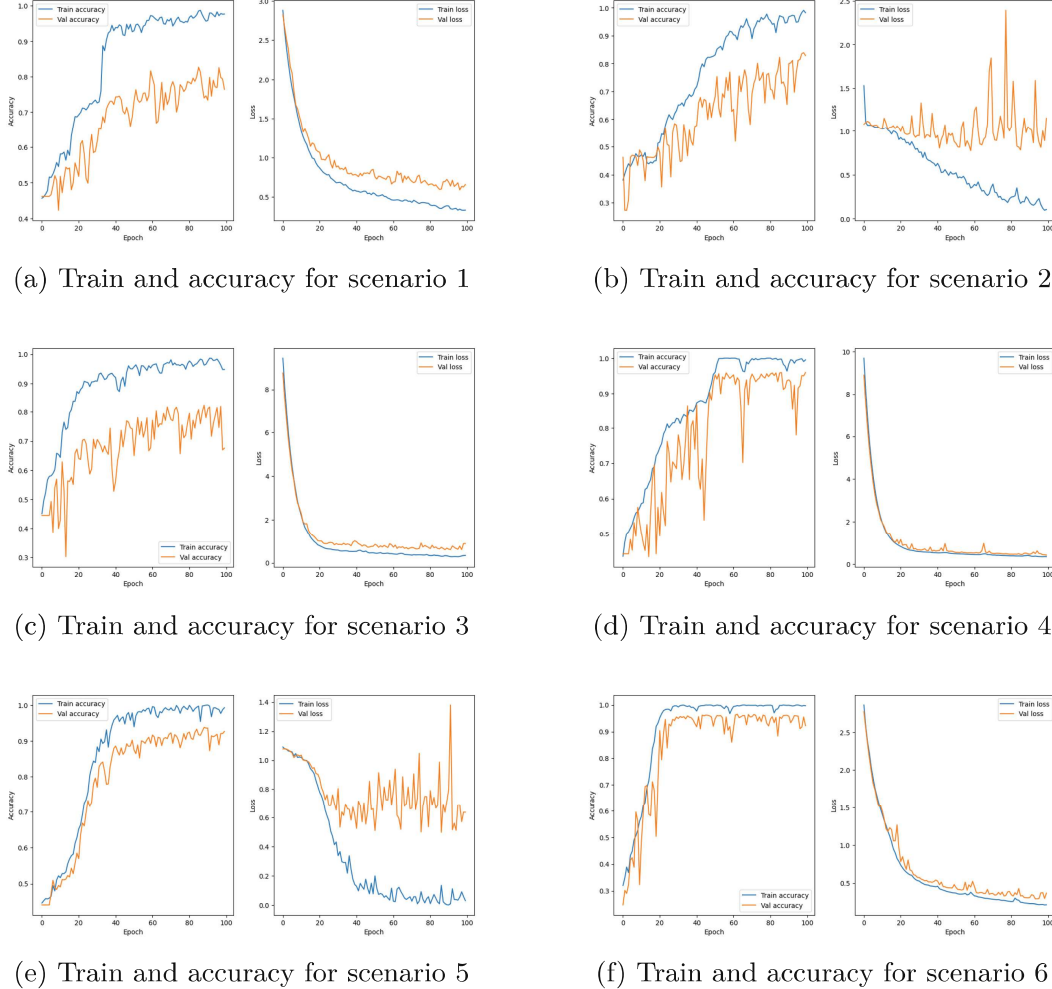


Fig. 5. The chart compares the accuracy and loss of the 6 scenarios.

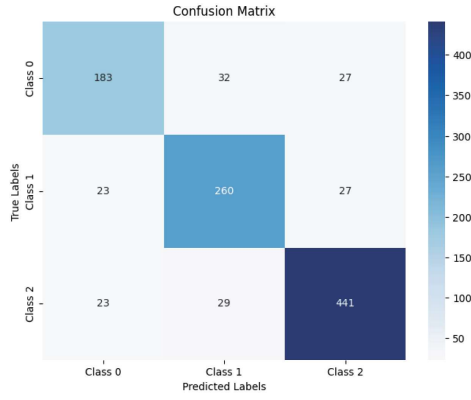
Based on the training results of 6 scenarios with loss of 1.3246, 0.8874, 0.6395, 0.5953, 0.387, 0.4110, we can see that the stability during training on scenarios 3 and 6 in 2 models is better than the remaining scenarios. For 2 scenarios using fuzzy jrip algorithm, the fuzzy rule set loses some important information on the image, leading to a large loss of information affecting the training and classification process. This shows the importance of the fuzzy rule set used in the training process will affect the output of the model as well as the classification results (Table 4).

Table 4. Model Performance Metrics

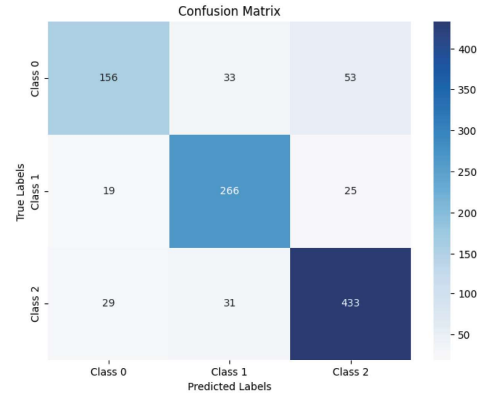
Model	Class	Precision (%)	Recall (%)	F1-score (%)	Accuracy (%)
ResNet50	Benign	79.91	75.62	77.71	84.59
	Malignant	81.00	83.87	82.41	
	Normal	89.09	89.45	89.27	
ResNet50_JRIP	Benign	76.47	64.46	69.69	81.82
	Malignant	80.61	85.81	83.12	
	Normal	84.74	87.83	86.25	
ResNet50_PART	Benign	81.74	73.97	77.66	85.55
	Malignant	82.69	89.35	85.89	
	Normal	88.91	88.84	88.88	
VGG19	Benign	92.44	90.91	91.67	93.68
	Malignant	90.74	94.84	92.74	
	Normal	96.27	94.32	95.29	
VGG19_JRIP	Benign	87.55	90.08	88.79	91.87
	Malignant	92.04	93.23	92.63	
	Normal	93.98	91.89	92.93	
VGG19_PART	Benign	91.97	94.63	93.28	94.83
	Malignant	95.11	94.19	94.65	
	Normal	96.11	95.33	95.72	

The table compares the performance of the ResNet50 and VGG19 models, along with the combination of the JRIP and PART algorithms, on three data classes: benign, malignant, and normal. In the ResNet50 group, the model using the fuzzy PART algorithm achieved the best performance with an overall accuracy of 85.55%, an improvement over the original model (84.59%) and JRIP (81.82%), especially in recognizing the Malignant class. However, when using the VGG19 network model, it showed complete superiority when the original model achieved accuracy 93.68%, with very good results in all data classes. In the VGG19 scenarios, VGG19_PART was the most effective model with the highest Accuracy of 94.83%, with the precision, recall, and F1-score indexes being equal in all three classes. Overall, VGG19_PART is the best choice due to its stable performance and outstanding accuracy, suitable for application in this classification problem (Fig. 6).

Based on the confusion matrix of 6 scenarios classified into 3 classes (0: Benign, 1: Malignant, 2: Normal), it can be observed that the classification performance of the models is quite different between the scenarios and the results are better when using the VGG19 network model. In class 2 (Normal), which often achieves the highest accuracy in most scenarios, this class has a superior correct prediction rate, such as in scenario 6, it reaches 490/513 correct predictions. However, confusion between classes 0: Benign and 1: Malignant appears



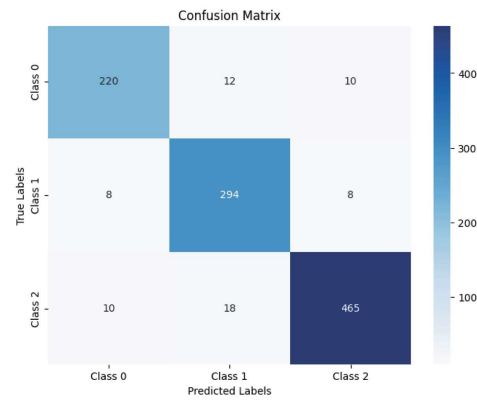
(a) Train and accuracy for scenario 1



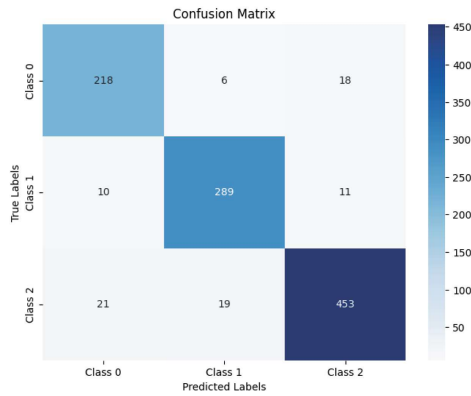
(b) Train and accuracy for scenario 2



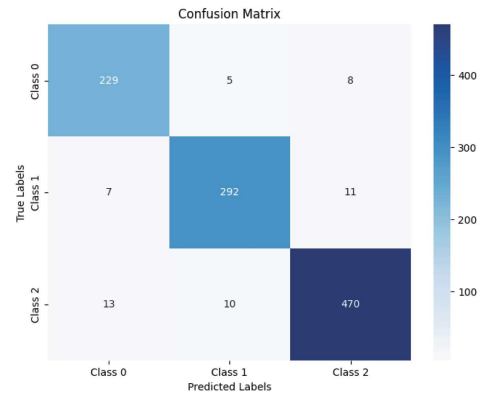
(c) Train and accuracy for scenario 3



(d) Train and accuracy for scenario 4



(e) Train and accuracy for scenario 5



(f) Train and accuracy for scenario 6

Fig. 6. Confusion matrix of the 6 scenarios.

frequently, especially in scenarios 1 and 2, because these are two classes with similar characteristics because in reality, the data in these 2 classes is often less than that in the Normal class, which also limits the generalization ability of the

model. Scenarios 4 and 6 show better performance in class 0, with high number of correct predictions (220/230 and 239/234 respectively). Class 1: Malignant improves significantly in scenarios 3 and 5, with accuracy of 277/313 and 279/313 respectively. Scenario 6 is considered the best with stable performance across all three classes (Fig. 7).

Benign	Actual: Benign Predicted: Benign (1.00)	Actual: Benign Predicted: Benign (0.91)	Actual: Benign Predicted: Benign (1.00)	Actual: Benign Predicted: Benign (1.00)
	(a)	(b)	(c)	(d)
	Actual: Malignant Predicted: Malignant (0.87)	Actual: Malignant Predicted: Malignant (0.79)	Actual: Malignant Predicted: Malignant (0.82)	Actual: Malignant Predicted: Malignant (0.85)
	(e)	(f)	(g)	(h)
Normal	Actual: Normal Predicted: Normal (1.00)	Actual: Normal Predicted: Normal (0.98)	Actual: Normal Predicted: Normal (0.98)	Actual: Normal Predicted: Normal (0.92)
	(i)	(j)	(k)	(l)

Fig. 7. The classification results on the test set belong to 3 categories: benign (a,b,c,d), malignant (e,f,g,h) and normal (i,j,k,l).

5 Conclusion

In the paper, we propose a classification model based on deep learning and fuzzy logic techniques. Through feature extraction using machine learning models, fuzzy logic algorithms are applied to eliminate unimportant features, thereby helping the classification model achieve high performance. Experimental results demonstrate that the combined model VGG19_PART achieves an accuracy of 94.83%. In addition, some combined algorithms have not achieved high performance due to the generality of the problem performed on many layers and the

limited combination with rule-generating algorithms. In the upcoming work, we will compare and improve the rule-generating models and combine additional features such as microcalcifications and tumors on the image to improve the accuracy of the model.

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